Bhavdeep Arora

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Education

Toronto Metropolitan University — *Toronto, ON*
Bachelor of Science (Honours) Computer Science · Software Engineering Concentration — *July 2026*
- Relevant Coursework: Data Structures & Algorithms, Software Engineering, Operating Systems, Data Science

Leadership & Volunteer Experience

Data for Good Toronto | *Data Analyst* — *May 2026 – Present*
- Collaborate with a cross-functional team of analysts, engineers, and stakeholders to evaluate datasets for non-profit and social impact organizations, supporting evidence-based decision making
- Develop data-cleaning, transformation, and exploratory analysis workflows in Python and pandas, contributing to end-to-end analytics pipelines for community-focused projects

SciXchange | *Technical Assistant* — *September 2023 – April 2024*
- Guided 50+ visitors through interactive science exhibits, translating technical STEM concepts into engaging explanations, while collaborating with a team of volunteers and staff to plan and run activities under busy, time-sensitive conditions

Projects & Research

bhavdeeparora.dev/projects

Thunderhead — Reverse Proxy for Automated Bot Detection | *Go, HTTP, Behavioral Analysis* | Live | Github
- Engineered a lightweight reverse proxy in Go that analyzes inbound HTTP traffic across **five behavioral heuristics**—including crawl velocity, robots.txt violations, request frequency, and header anomalies—to identify automated bot activity **without third-party security services**
- Developed a configurable **risk-scoring engine** that assigns request scores from 0–100 across **12** weighted signals and applies graduated mitigations (allow, tarpit, or block), correctly flagging **94%** of automated traffic across **50,000** requests while maintaining a **2%** false-positive rate on legitimate users

Precursor — Cross-Asset Momentum Spillover Research | *Python, statsmodels, VAR, NetworkX* | Paper | Github
- Ran an empirical study testing whether commodity futures momentum **Granger-causes** momentum in related sector equity ETFs, using daily data from 2010–2026 across **five commodity-equity pairs**
- Confirmed **all five** hypothesised spillovers at $\alpha = 0.05$ (Gold→GDX: F=18.90, p<0.001) via Granger causality, VAR, and DAG analysis; used rolling Granger tests, Chow structural break analysis, and a signal-based backtest achieving a **Sharpe ratio of 1.4** to show spillovers are regime-dependent rather than persistent

Groat — Self-Hosted LLM Cost Optimization Proxy | *Rust, Axum, SQLite, LanceDB* | Live | Github
- Built a **self-hosted, OpenAI-compatible LLM proxy** in Rust (4-crate workspace: server, core logic, provider layer, storage) that cuts API spend **without requiring any application code changes**
- Implemented **semantic response caching** via bge-small-en-v1.5 embeddings and LanceDB similarity search (0.95 threshold), automatic prompt-cache-boundary injection to claim provider-side **50–90%** cached-token discounts, and user-defined cost-tier routing, cutting redundant API spend by **31%** across **8,200** logged requests

Funes — Local-First Semantic Memory System | *Rust, Ollama, SQLite, CLI* | Live | Github
- Built a **local-first semantic memory system** in Rust that indexes files, notes, and shell history into vector embeddings, enabling natural-language retrieval across a developer's knowledge base **without cloud infrastructure**
- Architected a background indexing daemon with filesystem monitoring, chunking pipelines, and SQLite-backed vector storage; integrated local embedding and generation models to return relevant results in **85ms** on average across **3,400** indexed documents, achieving **91%** top-5 retrieval accuracy

Lacunae — Deep Learning MRI Reconstruction | *Python, PyTorch, U-Net, fastMRI* | Github
- Trained a **7.7M-parameter U-Net** on ∼100 GB of fastMRI single-coil knee MRI data to reconstruct **diagnostic-quality images** from 8× undersampled k-space, achieving **SSIM: 0.7803** and **PSNR: 30.8 dB** on held-out validation scans—an improvement of **18%** SSIM over zero-filled baseline reconstruction

Verrere — Swipe-Based Book Discovery Platform | *Next.js 15, TypeScript, Prisma, PostgreSQL* | Live | Github
- Built a full-stack book discovery platform delivering **personalized recommendations** via a swipe-based interface, integrating the Hardcover GraphQL API with PostgreSQL/Prisma persistence and achieving a **38%** swipe-to-save rate across **120** active users

Technical Skills

Languages: Python, Java, C/C++, JavaScript, TypeScript, SQL, Go, Rust, HTML/CSS, ASP.NET
Frameworks: React, Next.js, Node.js, Express, Django, Flask, Tailwind CSS, GraphQL
Developer Tools: Git, Docker, CI/CD, GitHub Actions, Linux, REST APIs, Postman, Nginx
Databases & Cloud: MySQL, PostgreSQL, MongoDB, SQLite, Redis, AWS, Azure, GCP, Firebase, Vercel

Certifications

AWS Certified Cloud Practitioner | *Amazon Web Services* — *August 2026*
- *Built a foundation in cloud architecture, security, and cost-efficient deployment*

Google Data Analytics Professional Certificate | *Google* — *August 2026*